

# Multi-Page Visually Rich Document Understanding

A survey of system design and an empirical attribution of failure

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- Introduction
- Survey of MP-VRDU
- Empirical Attribution Study
- Conclusion
- Proof of Concept: A Simple Method

# Multi-Page Visually Rich Documents

Documents carry meaning through **text**, **visual information** and **layout** together. **Multi-page** documents add another layer: meaning can run across pages, as when a sentence refers to a figure several pages away.

Figure 10: Consequences of Genetic Impairment Defects

Figure 10: Consequences of Genetic Impairment Defects. The diagram illustrates the flow of genetic information from DNA to the phenotype. It shows a flowchart of genetic inheritance, starting with a DNA double helix, leading to a cell, then to a protein, and finally to a phenotype. A photograph of a person's face is included, showing a genetic defect. The text explains that genetic defects can lead to various consequences, including developmental delays, physical abnormalities, and mental health issues.

Research paper  
born-digital

Define your problem

Page # 1

Redacted by DOI

Handwritten form with a redacted section. The form is titled "Define your problem" and contains several checkboxes and text fields. A large section of the form is redacted with a black box. The text "Redacted by DOI" is written in red ink over the redacted area. The form is dated "Page # 1" and "0252".

Signed form  
scanned, handwritten

Parts and Features

Below are illustrations of your appliance with the parts and features called out. Your model may have all of the features shown and it may not be exactly as illustrated.

Style 1

Technical manual showing a refrigerator with parts and features labeled. The diagram includes labels for various components such as the control panel, light, EZ-Track, model and serial number label, utility compartment, wire rack, egg bin, removable freezer shelves, snack bin, slide-out freezer baskets, leveling rollers, base grille, most drawer air control, crispers, humidity-controlled crisper, door bin slides, deep door bins, and adjustable door bins. The diagram is titled "Parts and Features" and "Style 1".

Technical manual  
layout-dependent

Table of Contents

Section Financial Data

The following selected consolidated financial data is not necessarily indicative of results of future operations and should be read in conjunction with Item 7, Management's Discussion and Analysis of Financial Condition and Results of Operations and Item 8, Financial Statements and Supplementary Data. The following amounts related to earnings per share and share outstanding have been adjusted for the Stock Split for all periods reported, see Note 8 of Item 8, Financial Statements and Supplementary Data for further detail on the Stock Split.

Consolidated Statements of Operations:

|                                            | 2012         | 2011         | 2010         | 2009         | 2008         |
|--------------------------------------------|--------------|--------------|--------------|--------------|--------------|
| Revenues                                   | \$ 6,779,311 | \$ 5,566,618 | \$ 4,374,342 | \$ 3,489,282 | \$ 3,294,377 |
| Operating income                           | 303,626      | 402,448      | 228,347      | 419,907      | 276,946      |
| Net income                                 | 127,641      | 266,799      | 112,493      | 173,152      | 226,126      |
| Earnings per share:                        |              |              |              |              |              |
| Basic                                      | \$ 0.29      | \$ 0.61      | \$ 0.28      | \$ 0.54      | \$ 0.61      |
| Diluted                                    | \$ 0.28      | \$ 0.62      | \$ 0.26      | \$ 0.64      | \$ 0.59      |
| Weighted average common shares outstanding | 437,889      | 435,144      | 407,383      | 318,648      | 369,919      |
| Diluted                                    | 436,416      | 433,894      | 407,817      | 412,327      | 369,183      |

Consolidated Statements of Cash Flows:

|                                                     | 2012         | 2011      | 2010      | 2009      | 2008       |
|-----------------------------------------------------|--------------|-----------|-----------|-----------|------------|
| Net cash (used in) provided by operating activities | \$ (769,439) | \$ 36,483 | \$ 97,831 | \$ 21,586 | \$ 317,732 |
| Free cash flow (1)                                  | (926,517)    | (226,689) | (26,340)  | (38,315)  | 186,519    |

(1) See "Liquidity and Capital Resources" for a definition of "free cash flow" and a reconciliation of "free cash flow" to "net cash (used in) provided by operating activities."

Consolidated Balance Sheets:

|                                                   | 2012         | 2011         | 2010         | 2009        | 2008        |
|---------------------------------------------------|--------------|--------------|--------------|-------------|-------------|
| Cash, cash equivalents and short-term investments | \$ 2,310,715 | \$ 1,808,496 | \$ 1,200,491 | \$ 748,078  | \$ 797,811  |
| Total current assets, net (2)                     | 3,718,313    | 4,519,405    | 3,535,564    | 2,814,089   | 2,522,619   |
| Working capital (3)                               | 1,802,216    | 1,263,889    | 881,049      | 553,887     | 595,780     |
| Total assets (3)                                  | \$2,362,871  | \$2,042,269  | \$1,404,023  | \$1,041,781 | \$1,062,111 |
| Long-term debt (2)                                | 2,273,562    | 881,889      | 495,462      | 395,782     | 395,223     |
| Long-term debt due to related party (1)           | —            | —            | —            | 198,209     | 197,792     |
| Total current liabilities                         | 2,626,340    | 1,775,082    | 1,341,560    | 1,076,422   | 799,632     |
| Total liabilities                                 | 4,811,383    | 3,660,973    | 3,212,573    | 2,465,409   | 1,674,664   |
| Total assets less liabilities                     | 2,225,438    | 1,807,796    | 1,032,941    | 744,475     | 642,830     |

(1) Certain prior year amounts have been reclassified to conform to the current year presentation in the consolidated financial statements. See Note 1 and Note 2 in Item 8, Financial Statements and Supplementary Data for further detail.

Financial report  
dense tables

Leave a lasting impression.

Product brochure for the Camry SE Hybrid. The brochure features a large photograph of the car and several smaller images showing interior details. The text highlights the car's design, performance, and features, including the "Leave a lasting impression" headline. The brochure is titled "Camry SE Hybrid" and "Leave a lasting impression."

Product brochure  
photo-led

- **Text.** Typeset or handwritten, embedded in the file or only visible in a scan.
- **Visual information.** Charts, figures, photographs and handwriting.
- **Layout.** How information is arranged in the two-dimensional space of a page.

# What is Document Understanding?

Answering natural-language questions: **identify the relevant content** and **connect the evidence**, within or across pages.

Long documents make this **very difficult**: evidence is sparse, spread across many pages, and often more than a model can read at once.

## LAYOUT ANALYSIS

3

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Latino

Increasingly Confident in Personal Finances, See Better Economic Times Ahead

Yet many economic indicators show few gains for the community since the Great Recession

Latinos have become considerably more upbeat about their personal finances and optimistic about their financial future since the Great Recession, according to newly released results from a national survey of Latino adults. The survey also shows that Latinos have pulled even with the general U.S. population in their views of their personal finances and continue to outpace them on optimism about the future. However, community economic indicators show limited progress since the Great Recession.

Four-in-ten Latinos say their personal finances are in "excellent" or "good" shape, a 17 percentage point increase since 2008, when only 23% made a similarly positive assessment of their finances (the Great Recession began in December 2007). By contrast, the share of all Americans who have a similarly positive view of their finances remained essentially flat during the same seven-year period (45% in 2008 vs. 43% in 2015).

The survey also shows that Hispanics are more upbeat in their financial expectations for the upcoming year than they were in 2008. About eight-in-ten Hispanic adults (81%) say they expect their family's financial situation to improve in the next year, up from 67% who said the same in 2008. By comparison, the U.S. public is not as upbeat –

Hispanics more confident in personal finances than in 2008, see better economic times ahead

When asked about their personal financial situation, 76% of Hispanics expect it to improve "a lot" or "some" in the next year, up from 67% in 2008. By comparison, the U.S. public is not as upbeat – 70% expect improvement.

| Year | Hispanic Confidence (%) | Hispanic Expectation (%) | General Confidence (%) | General Expectation (%) |
|------|-------------------------|--------------------------|------------------------|-------------------------|
| 2008 | 23                      | 67                       | 45                     | 67                      |
| 2016 | 76                      | 58                       | 43                     | 61                      |

Source: National Survey of Latinos conducted Oct. 23-Nov. 30, 2015; survey conducted Dec. 9-12, 2015; Oct. 24-Nov. 20, 2012; Dec. 12-16, 2012; Dec. 7-12, 2011; Dec. 3-8, 2008; Dec. 9-12, 2008; Aug. 9-10, 2004.

Latino: National Survey of Latinos in Personal Finances, See Better Economic Times Ahead

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## DOCUMENT OUTLINE

### Research paper, 24 pages

5.2 Conscious Incompetence

Paragraphs

Figure 4

Caption

5.3 Retrieval Analysis

Paragraphs

Figure 5

Caption

5.4 Human Evaluation

Paragraphs

Table 6

Caption

### EXAMPLE QUESTION

“Does the model rated best by humans in Table 6 also have the best citation recall in Table 3?”

Evidence needed:

Table 6, this page

Table 3, previous page

**Cross-page:** it compares two tables on different pages, so neither alone is enough.

■ Title   ■ Section heading   ■ Text   ■ Figure / chart   ■ Table   ■ Caption / note   ■ Header / footer

Background   Survey   Empirical   04 / 25

# LLM Assistants

How often does the final pipeline stop at its earliest allowed point, and how does its page usage compare with the number of gold evidence pages?

Read a file, updated tasks ▾

Added task Locate pages for Figure 4.2 and Table 4.2 ▾

Request  
Description  
Map printed page numbers to PDF pages using the contents list and the offset seen on page 27

Added task Read figure and table pages visually ▾

Request  
Description Read PDF pages 60-61 and extract the bar heights and token statistics

Added task Verify and answer ▾

Request  
Description Cross-check figure values against surrounding text, then answer

Completed task Locate pages for Figure 4.2 and Table 4.2

Started task Read figure and table pages visually

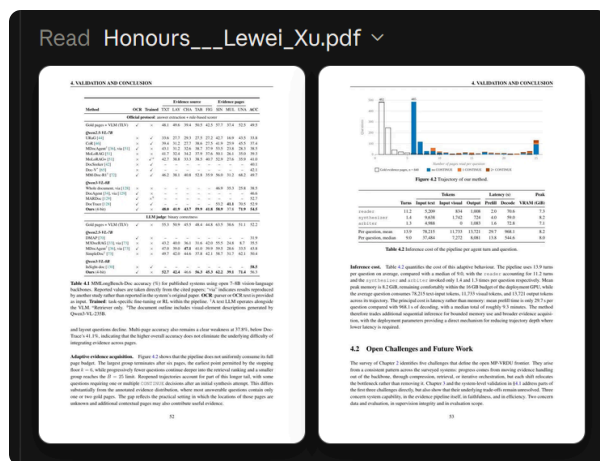
Read Honours\_\_\_Lewei\_Xu.pdf ▸

Verifying the data points plotted across the chart.

Completed task Read figure and table pages visually

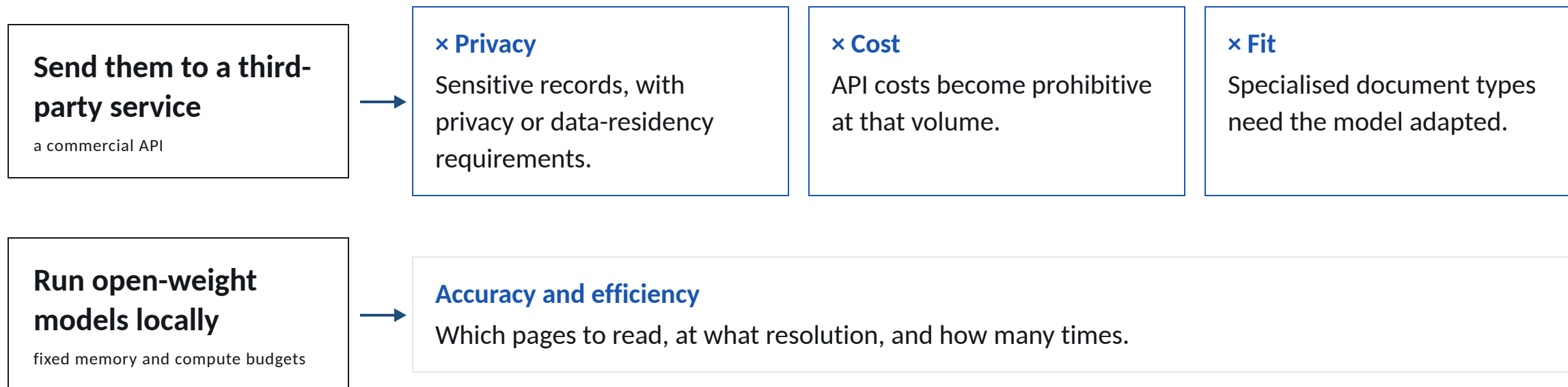
Completed task Verify and answer

- Upload a document to an assistant and **ask it questions**.
- My thesis, almost **100 pages**, and a difficult question: it still did well.
- An **agentic workflow**: it plans, locates the relevant pages, and inspects them before answering.
- It **renders pages as images**, reading visually as well as through text.
- Research reaching consumer products, but only one part of how document understanding is used in practice.



# Real-World Constraints

In industry: classify, summarise or extract information from **thousands, or even millions, of documents**.



The central question: how can we improve document understanding **under a fixed compute budget**?

# Table of Contents

- Introduction

- **Survey of MP-VRDU**

*Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding.*

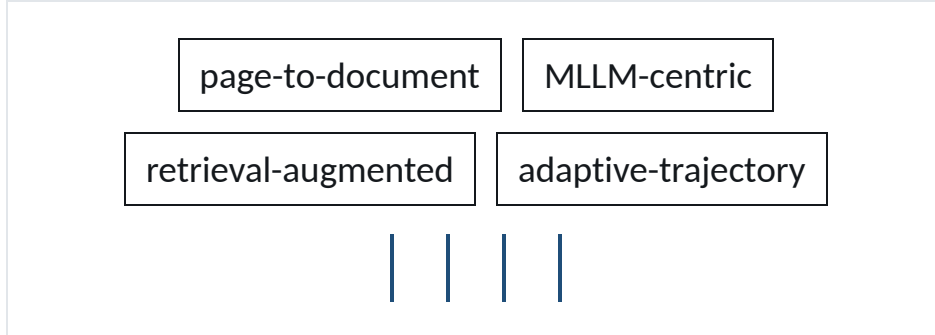
Lewei Xu, Yihao Ding, Zihan Xu, Siwen Luo, Daochang Liu, Wei Liu

[EMNLP 2026, main conference](#)

- Empirical Attribution Study
- Conclusion
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# Survey: Research Gap and Contributions

## RESEARCH GAP



Multi-page systems developed along **parallel lines**, with different terminology and little comparison between them.

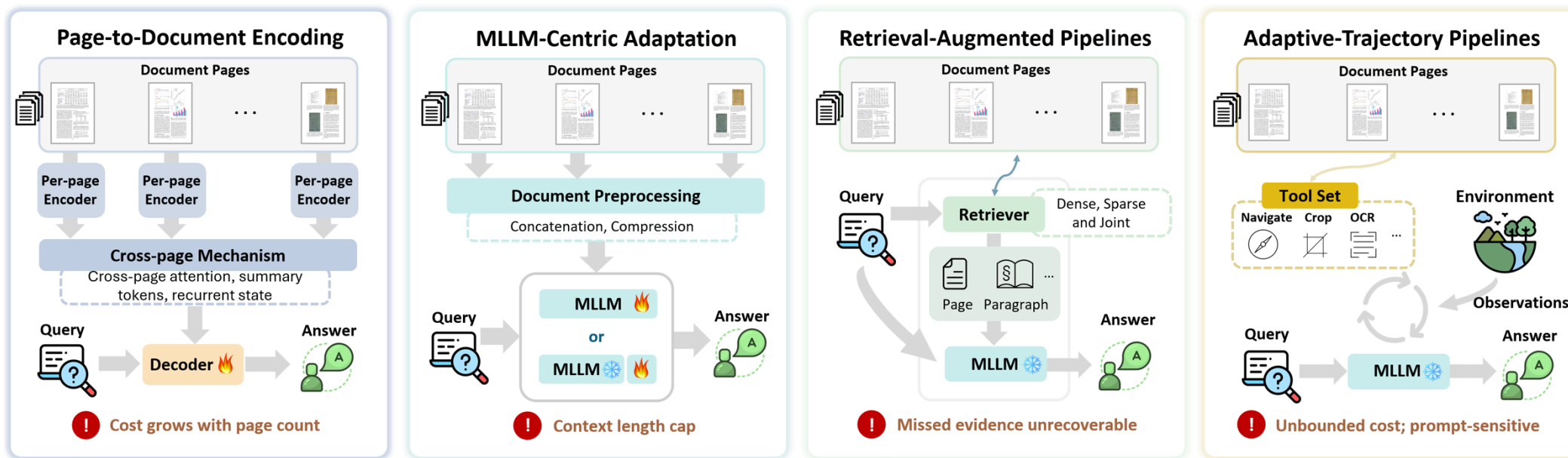
## CONTRIBUTIONS

- 1 Brings these systems together through **evidence management**: how a system manages the relevant evidence across a very long document.
- 2 Organises them into **four architectural families**.



# Multi-Page Architectural Overview

Four architectural families. They emerged roughly in this order, although their development overlaps.



**Late 2023 to mid 2024**

Encode each page, connect them with a cross-page mechanism, train end to end

**Early 2024 to mid 2026**

Feed all pages into a pretrained model's context, with further training

**Early 2024 to mid 2026**

Retrieve evidence first, then answer.  
The workflow is fixed in advance

**Early 2025 to mid 2026**

A language model decides what to inspect next from what it has found

Snowflake indicates typically frozen, fire indicates typically trained

# Surveyed Systems at a Glance

| Model                            | Venue       | OCR  | Vision Encoder          | LLM Backbone                   | Mod.    | Search         | Reasoning   | Agentic    |
|----------------------------------|-------------|------|-------------------------|--------------------------------|---------|----------------|-------------|------------|
| Page-to-Document Encoding Models |             |      |                         |                                |         |                |             |            |
| Arctic-TILT (2025)               | ACL         | Yes  | U-Net                   | T5-Large                       | T, V, L | -              | -           | -          |
| GRAM (2024)                      | CVPR        | Yes  | DocFormerv2             | DocFormerv2                    | T, V, L | -              | -           | -          |
| Hi-VT5 (2023)                    | PR          | Yes  | DiT                     | T5-Base                        | T, V, L | -              | -           | -          |
| RM-T5 (2024)                     | DAS         | Yes  | DiT                     | T5-Base                        | T, V, L | -              | -           | -          |
| MLLM-Centric Adaptation Models   |             |      |                         |                                |         |                |             |            |
| DocR1 (2026)                     | AAAI        | No   | Qwen2.5-VL              | Qwen2.5-VL-7B                  | V       | -              | CoT, HD     | -          |
| DocSeeker (2025)                 | CVPR        | No   | Qwen2.5-VL ViT          | Qwen2.5-VL-7B-Instruct         | V       | -              | CoT, HD     | -          |
| Leopard (2024)                   | TMLR        | No   | SigLIP-SO400M           | LLaMa-3.1-8B, Mistral-7B       | V       | -              | CoT         | -          |
| mPLUG-DocOwl2 (2025)             | ACL         | No   | ViT-L/14                | mPLUG-Owl2                     | V       | -              | -           | -          |
| URaG (2026)                      | AAAI        | No   | Qwen2.5-VL ViT          | Qwen2.5-VL-3B / Qwen2.5-VL-7B  | V       | -              | HD          | -          |
| Docopilot (2025b)                | CVPR        | No*  | InternViT-300M          | Multiple (Intern)              | V       | -              | -           | -          |
| Texthawk2 (2024)                 | preprint    | No*  | SigLIP-SO400M           | Qwen2-7B-Instruct              | V       | -              | -           | -          |
| DocSLM (2025)                    | CVPR        | Yes  | SigLIP2                 | Qwen2.5-1.5B                   | T, V, L | -              | -           | -          |
| InstructDoc (2024)               | AAAI        | Yes  | CLIP                    | FlanT5, BLIP2                  | T, V, L | -              | -           | -          |
| LayTokenLLM (2025b)              | CVPR        | Yes  | -                       | LLaMA3-8B, Qwen1.5-7B          | T, L    | -              | -           | -          |
| CoR (2026)                       | preprint    | Yes* | Qwen2.5-VL              | Qwen2.5-VL-7B                  | T, V    | -              | CoT, PE, HD | -          |
| Retrieval-Augmented Pipelines    |             |      |                         |                                |         |                |             |            |
| AVIR (2025)                      | MMAAsia     | No   | Pix2Struct              | Qwen2.5-VL-3B-AWQ              | V       | D              | -           | -          |
| DFVC (2026)                      | Electronics | No   | VisRAG, ColQwen         | Qwen2-VL-2B-Instruct           | V       | D              | -           | -          |
| DREAM (2025)                     | MM          | No   | InternVL2 ViT           | InternVL2-4B                   | V       | J              | HD          | -          |
| M3DocRAG (2024)                  | preprint    | No   | ColPali, ColQwen        | Multiple (Qwen/Intern/Idefics) | V       | D              | -           | -          |
| MM-R5 (2026)                     | AAAI        | No   | Qwen2.5-VL ViT          | Qwen2.5-VL-7B                  | V       | D              | CoT         | -          |
| MoLoRAG (2025b)                  | EMNLP       | No   | ColPali                 | Multiple (Qwen/DS/LLaVA)       | V       | D, Nav         | -           | 1A         |
| Self-Attention Scoring (2024)    | ICDAR       | No   | Pix2Struct              | Pix2Struct                     | V       | D              | -           | -          |
| SLEUTH (2025)                    | preprint    | No   | ColPali, Qwen3-VL ViT   | Multiple (Qwen/GLM/Gemini)     | V       | D              | HD          | 4A         |
| SV-RAG (2025)                    | ICLR        | No   | InternVL2, Phi-3-vision | Multiple (Intern/Pali/Phi)     | V       | D              | -           | -          |
| VDocRAG (2025)                   | CVPR        | No*  | Phi3V-4.2B              | Phi3V-4.2B                     | V       | D              | -           | -          |
| CREAM (2024)                     | MM          | Yes  | Pix2Struct              | LLaMA2-7B                      | T, V    | J              | HD          | -          |
| KGP (2024c)                      | AAAI        | Yes  | -                       | Multiple                       | T, L    | S, IR, Nav     | -           | 1A         |
| MHier-RAG (2025)                 | preprint    | Yes  | Qwen-VL-Plus            | Multiple (Qwen/DS/GPT)         | T, V, L | J              | CoT, HD     | -          |
| MLDocRAG (2026)                  | preprint    | Yes  | Qwen2.5-VL              | Multiple (Qwen)                | T, V, L | D              | CoT, HD     | -          |
| MultiDocFusion (2025)            | ACL         | Yes  | -                       | Multiple (Qwen/LLaMa/Mistral)  | T, V, L | D              | HD          | -          |
| M2RAG (2025a)                    | NeurIPS     | Yes* | VisRAG-Ret              | Multiple (Qwen)                | T, V    | J              | -           | 3A         |
| MDocAgent (2025)                 | preprint    | Yes* | ColPali, ColQwen2       | Multiple (Llama/Qwen)          | T, V    | J              | -           | 5A         |
| PDF-WuKong (2024)                | preprint    | Yes* | IXC2-VL-4KHD            | IXC2-VL-4KHD                   | T, V    | J              | -           | -          |
| RAG-DocVQA (2025)                | ICDAR       | Yes* | DiT, Pix2Struct         | Qwen2.5-VL-7B-Instruct         | T, V, L | D              | -           | -          |
| VisDoMRAG (2025)                 | NAACL       | Yes* | ColPali / ColQwen2      | Multiple(Qwen/Gemini/GPT)      | T, V    | J              | CoT         | 3A, SC     |
| Adaptive-Trajectory Pipelines    |             |      |                         |                                |         |                |             |            |
| Doc-V* (2026)                    | preprint    | No   | Qwen2.5-VL              | Qwen2.5-VL-7B-Instruct         | V       | D, IR, QR, Nav | ReAct, HD   | 1A         |
| DocReact (2025a)                 | ACL         | No   | ColPali, VisRAG         | GPT-4o                         | V       | D, IR, QR      | ReAct, QD   | 1A         |
| MACT (2025)                      | preprint    | No   | Multiple                | Multiple (Qwen/MiMo/Intern)    | V       | D              | CoT, PE     | 4A, SC, SV |
| VRAG-RL (2026)                   | NeurIPS     | No   | Qwen2.5-VL ViT          | Qwen2.5-VL-3B/7B-Instruct      | V       | D, IR, QR      | ReAct, HD   | 1A         |
| DocLens (2025a)                  | preprint    | Yes  | -                       | Multiple (GPT/Gemini/Claude)   | T, V, L | J              | CoT         | 4A, SA     |
| DMAP (2026)                      | WWW         | Yes* | ColPali                 | GPT-4o / Qwen-plus             | T, V, L | J              | HD          | 2A, SR     |
| DocAgent (2025)                  | EMNLP       | Yes* | -                       | Multiple (GPT/Gemini/Claude)   | T, V, L | J, IR, Nav     | ReAct, HD   | 3A         |
| DocDancer (2026)                 | preprint    | Yes* | Qwen3-VL-235B           | Qwen3-4B/30B-A3B               | T, V, L | J, IR, QR      | ReAct       | 1A         |
| MM-Doc-R1 (2026)                 | preprint    | Yes* | Qwen2.5-VL ViT          | Qwen3-4B/8B                    | T, V    | S, IR, QR, Nav | ReAct, HD   | 3A         |
| SimpleDoc (2025)                 | EMNLP       | Yes* | ColQwen-2.5             | Multiple (Qwen)                | T, V    | J, IR, QR      | -           | 1A, SR, SV |
| ViDoRAG (2025a)                  | EMNLP       | Yes* | NV-Embed-V2, ColQwen2   | Multiple (Qwen/LLaMa/GPT)      | T, V    | J, IR          | ReAct, HD   | 3A, SR, SV |

## 1 Models and pipelines

The first two families are **models**: trained end to end, with components built for that architecture. The last two are **pipelines**: components are often interchangeable, **especially the backbone**.

## 2 Modalities

Encoding models combine **text, vision and layout**. Models in the second family mostly use **vision only**.

## 3 Search

Only appears once **retrieval is separated from answering**. Retrieval-augmented pipelines add a retrieval step.

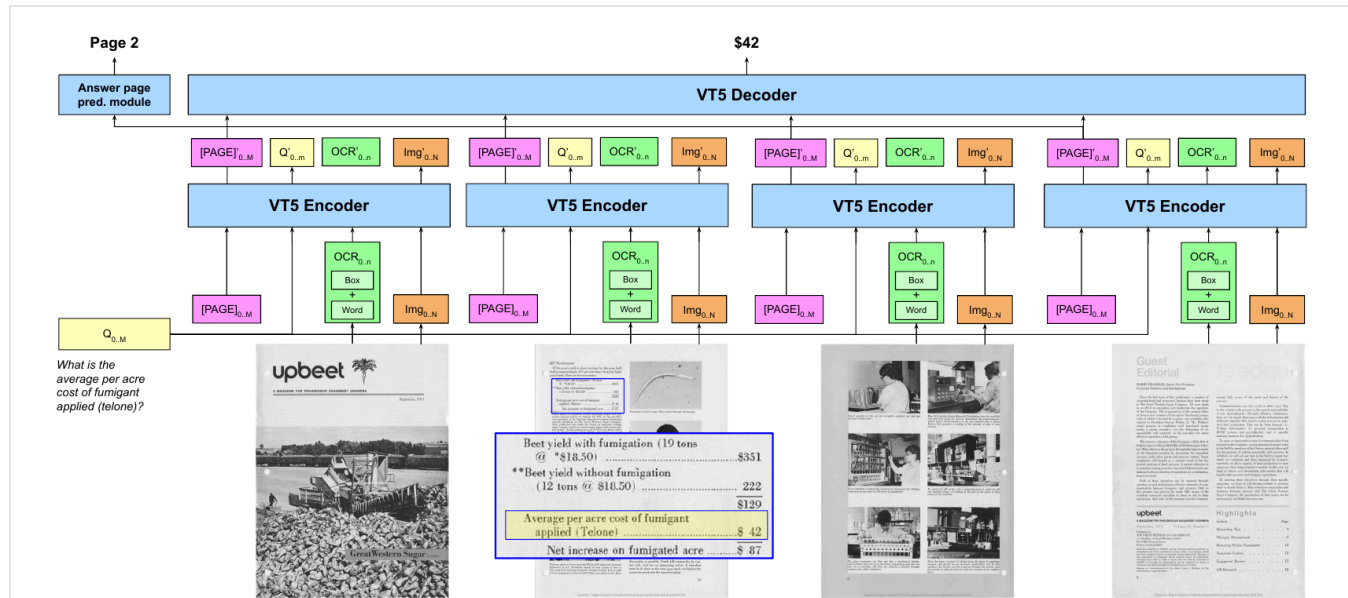
# Page-to-Document Encoding: Hi-VT5

Page-to-document encoding

MLLM-centric adaptation

Retrieval-augmented

Adaptive-trajectory



Hi-VT5 architecture. Each page is encoded into  $[PAGE]$  tokens, and the decoder reads the tokens of all pages to generate the answer. A second module predicts the answer page. From Tito et al., Pattern Recognition 2023.

## HOW IT WORKS

- Builds on **T5**, a text encoder-decoder, and adds visual and layout information.
- Each page is **encoded separately**, with the question, the recognised words and their positions, and patches of the page image.
- Ten learnable  **$[PAGE]$  tokens** summarise what on that page is relevant to the question.
- The decoder reads the **joined page tokens**, not the full pages, and generates the answer.

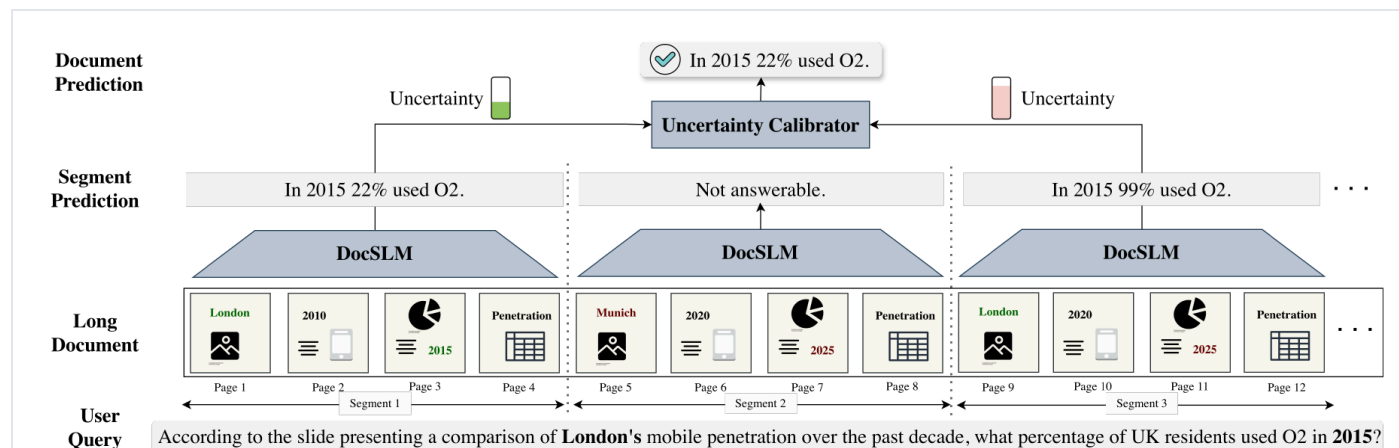
# MLLM-Centric Adaptation: DocSLM

Page-to-document encoding

MLLM-centric adaptation

Retrieval-augmented

Adaptive-trajectory



*Streaming abstention in DocSLM. The document is split into segments. Each segment gives an answer or abstains, and an uncertainty calibrator keeps the answer with the lowest uncertainty. From Hannan et al., CVPR 2026 Findings.*

## HOW IT WORKS

- A **small vision-language model** for long documents **under limited memory**.
- **Hierarchical multimodal compressor**. Each page's visual, text and layout information is jointly encoded into a **fixed 576 tokens**.
- **Streaming abstention**. The document is split into segments and processed one at a time. Each segment answers or abstains.
- An **uncertainty calibrator** keeps the answer with the lowest uncertainty.

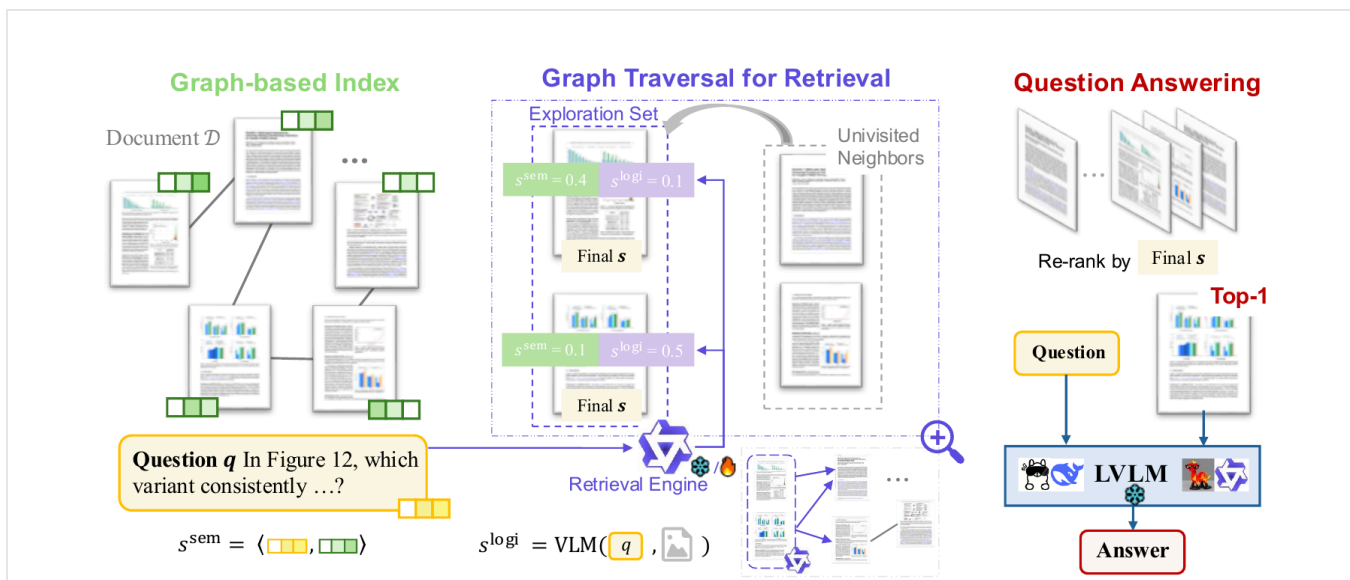
# Retrieval-Augmented: MoLoRAG

Page-to-document encoding

MLLM-centric adaptation

Retrieval-augmented

Adaptive-trajectory



The MoLoRAG framework: a graph-based index over pages, graph traversal for retrieval using semantic and logical relevance scores, then question answering over the top pages. From Wu et al., EMNLP 2025.

## HOW IT WORKS

- **Graph-based index.** A visual retriever represents each page, and pages are connected by their similarity.
- **Graph traversal.** The search starts from pages similar to the question. A small vision-language model helps score relevance, and the top K pages are kept.
- **Question answering.** The top K pages go to a separate vision-language model, the reasoner.

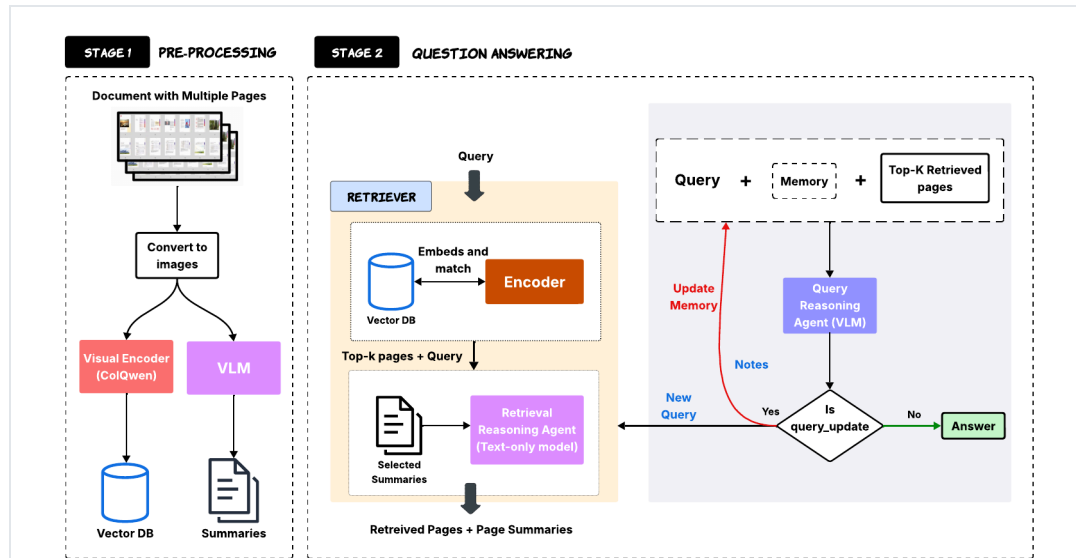
# Adaptive-Trajectory: SimpleDoc

Page-to-document encoding

MLLM-centric adaptation

Retrieval-augmented

Adaptive-trajectory



SimpleDoc. Stage 1 indexes each page as an embedding and a summary. Stage 2 retrieves and re-ranks pages, and a reasoning agent answers or updates its memory and query. From Jain et al., EMNLP 2025.

## HOW IT WORKS

- **Pre-processing.** Every page is indexed twice: as a **visual embedding**, and as a short **summary** written by a vision-language model.
- **Dual-cue retrieval.** Pages are shortlisted by embedding similarity, then a language model **re-ranks them using the summaries**.
- **Reasoning agent.** It reads the retrieved pages with a **working memory** of earlier notes, then either answers or **issues a refined query** to retrieve more.
- The loop ends when it answers, decides the question is not answerable, or reaches a step limit.

# Table of Contents

- Introduction
- Survey of MP-VRDU

- **Empirical Attribution Study**

*Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution*

Lewei Xu, Yihao Ding, Zihan Xu, Daniel Yitian Su, Daochang Liu, Siwen Luo, Yifan Peng, Wei Liu

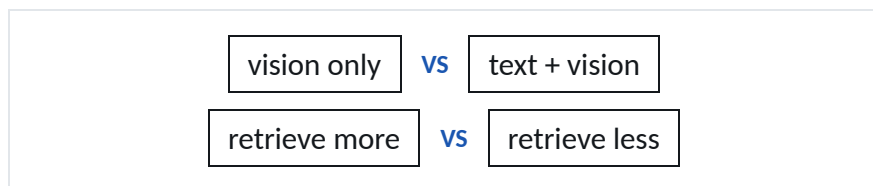
[Under review](#)

- Conclusion
- Proof of Concept: A Simple Method

# Empirical Study: Research Gaps and Contributions

## RESEARCH GAPS

- 1 **Competing design claims**, each tested on a different pipeline and dataset.



- 2 **No end-to-end empirical study** of where document understanding systems fail.

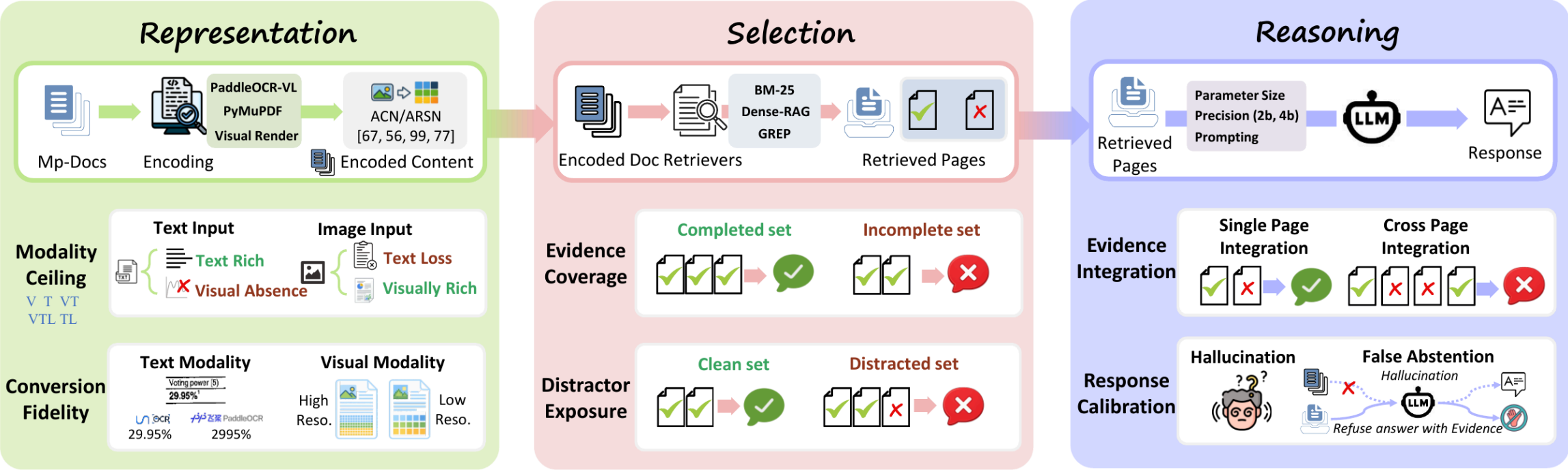
## CONTRIBUTIONS

- 1 A **unified view of failure**: three architecture-independent loci, representation, selection and reasoning.
- 2 Controlled interventions in **one pipeline**, across document collections and independently developed model families.
- 3 **Practical guidance** for designing and deploying systems **under a compute budget**, from qualitative examples and accuracy-cost comparisons.

*This is the local-deployment setting from the constraints slide.*



# Attribution Framework



## REPRESENTATION

Was the information preserved when the page was converted?

## SELECTION

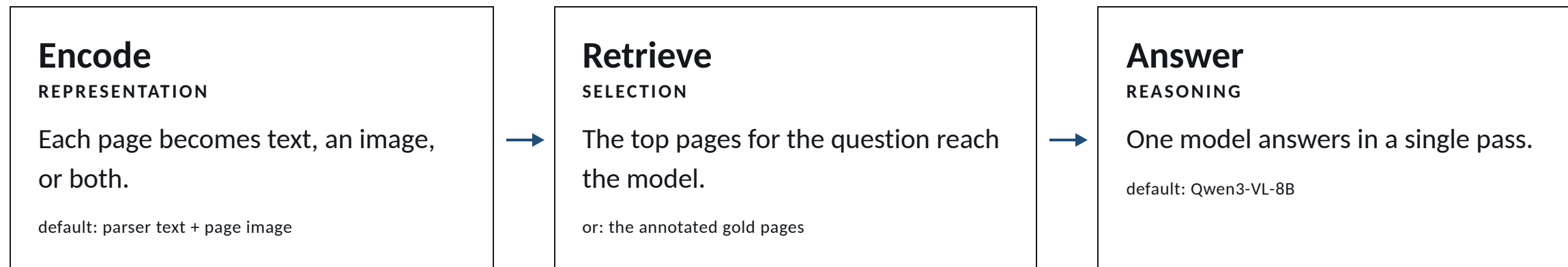
Did it actually reach the model?

## REASONING

Did the model use it correctly?

# Attribution by Construction

One simple pipeline. **Change one stage, hold the other two fixed.**



- **Gold pages** bypass retrieval, so selection can be taken out of the picture.
- **MMLongBench-Doc**: 135 documents, annotated with evidence pages, evidence type and answerability.

# Representation: Results

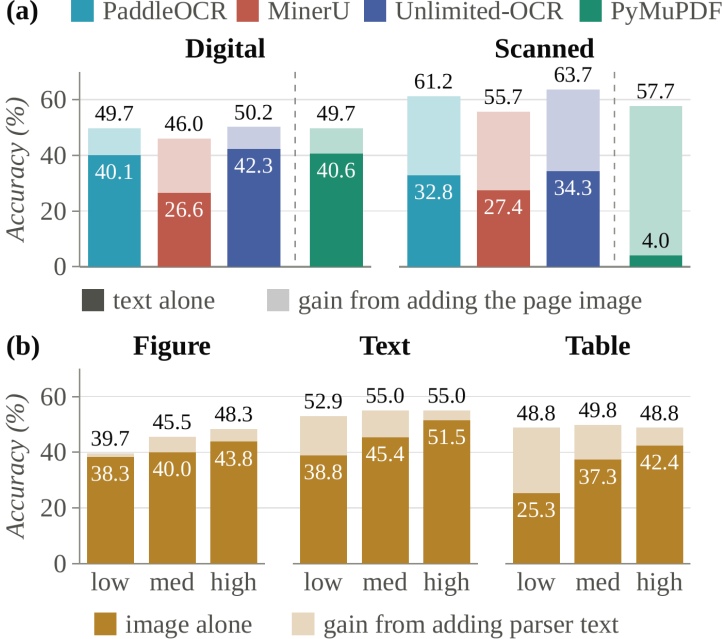
Is the evidence in the input?

## Modality ceiling

| Evidence               | Abstention  |             |            |            | Accuracy |             |             |      |
|------------------------|-------------|-------------|------------|------------|----------|-------------|-------------|------|
|                        | T           | TL          | TLV        | V          | T        | TL          | TLV         | V    |
| <b>MMLongBench-Doc</b> |             |             |            |            |          |             |             |      |
| Text                   | 23.0        | 11.5        | 7.3        | 8.0        | 36.9     | 44.4        | <b>51.0</b> | 42.3 |
| Layout                 | 20.3        | 10.2        | 0.8        | 4.2        | 26.3     | 33.9        | <b>49.2</b> | 43.2 |
| Table                  | 15.9        | 7.2         | 6.7        | <b>9.1</b> | 35.6     | <b>44.7</b> | <b>44.7</b> | 34.1 |
| Chart                  | 32.6        | <b>28.1</b> | 7.3        | 6.2        | 16.3     | 16.9        | <b>42.7</b> | 40.4 |
| Figure                 | <b>36.6</b> | 24.8        | <b>7.6</b> | 7.9        | 14.5     | 22.1        | <b>42.4</b> | 39.0 |
| All                    | 26.3        | 17.2        | 6.2        | 7.1        | 29.0     | 35.7        | <b>49.5</b> | 42.6 |
| <b>LongDocURL</b>      |             |             |            |            |          |             |             |      |
| Text                   | 4.6         | 3.5         | <b>1.5</b> | <b>1.6</b> | 71.8     | 75.6        | <b>78.2</b> | 70.0 |
| Layout                 | 2.1         | 1.6         | 0.7        | 0.9        | 45.0     | 49.2        | <b>51.3</b> | 49.3 |
| Table                  | 5.0         | 3.2         | 1.1        | 0.8        | 57.7     | <b>65.6</b> | 63.6        | 52.4 |
| Figure                 | <b>6.7</b>  | <b>8.6</b>  | <b>1.5</b> | 1.3        | 59.9     | 62.8        | <b>69.9</b> | 61.9 |
| All                    | 4.7         | 4.3         | 1.3        | 1.3        | 60.3     | 64.5        | <b>67.3</b> | 59.4 |

Text-only input refuses **chart and figure** questions most. Page images bring refusal **under 10%**, yet images alone trail text plus images.

## Conversion fidelity



Adding page images cuts the spread across parsers from **15.7 to 4.2 points**. Parser text makes resolution matter less.

# Representation: Design and Deployment

Is the evidence in the input?

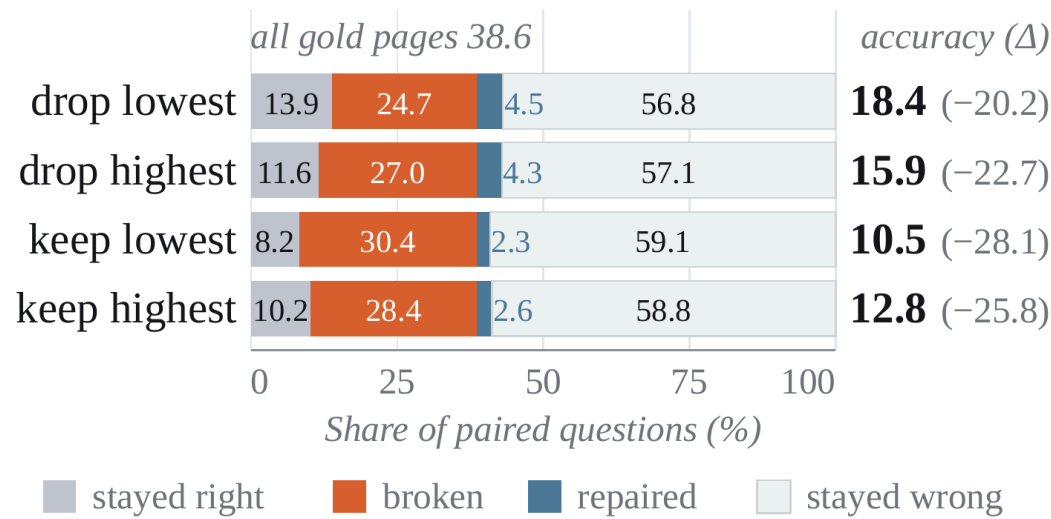
| Modality ceiling                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | two inline icons, parsed as <img> | Conversion fidelity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | scanned slide, text by parser |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| <p>Q What are the colors of the icons that the users touch to move the app up and remove the app respectively?</p> <p>Gold gray, red</p> <p>T <span>✗</span> "The document does not specify the colors"</p> <p>TL <span>✗</span> "it does not mention any visual details"</p> <p>--- ▼ add the page image -----</p> <p>V <span>✓</span> "The icon is gray. ... The icon is red."</p> <p>TLV <span>✓</span> "gray (a gray upward arrow) ... red"</p> <p>Colour exists only in pixels; a text input leaves the reasoner nothing to read.</p> |                                   | <p>Q What is the name of the company that created this presentation?</p> <p>Gold AVALAUNCH MEDIA</p> <p>PaddleOCR <span>✗</span> "Asterks" (parse: KICK ASTERKS)</p> <p>MinerU <span>✗</span> "does not provide any evidence"</p> <p>--- ▼ re-convert, or add the image -----</p> <p>Unlimited <span>✓</span> "AVALAUNCH MEDIA"</p> <p>with image <span>✓</span> "Avalaunch Media ... in the banner"</p> <p>The error sits in the transcript, not the reasoner, and each parser fails differently.</p> |                               |

- DESIGN
- Complementary modalities recover what one input loses: prioritise reliable embedded text, add images when appearance matters or extraction is unreliable, re-parse when the text is inadequate.
- DEPLOYMENT
- Embedded text plus images matches parser text plus images (50.1% against 49.5%) with fewer tokens. Images cost 26.3 s of prefill against 1.8 s, so set modality and selection depth together.

# Selection: Results

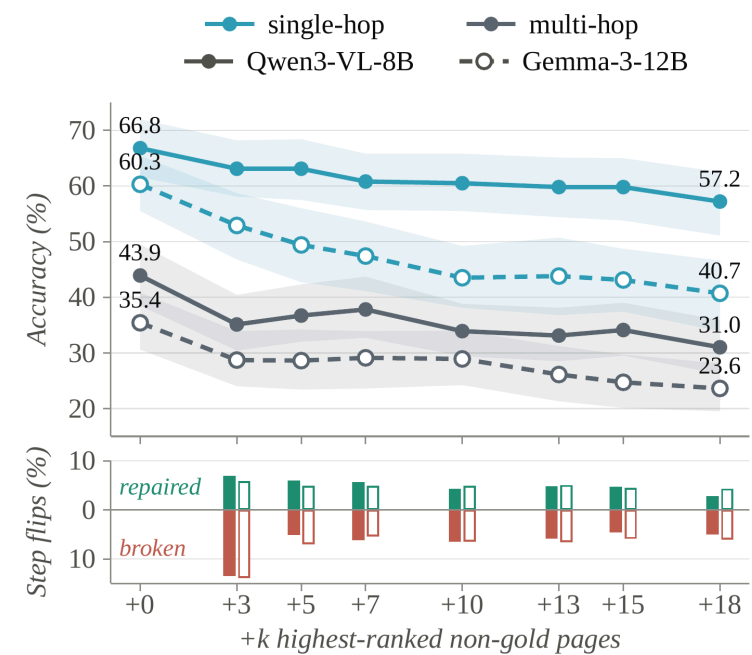
Is the right evidence in context?

## Evidence coverage



Removing **one gold page** takes accuracy from **38.6% to 18.4%** or 15.9%. Retrieval rank does not predict which page is safe to lose.

## Distractor exposure



With every gold page kept, three extra pages drop multi-hop accuracy from **43.9% to 35.1%**. Both backbones show it.

# Selection: Design and Deployment

Is the right evidence in context?

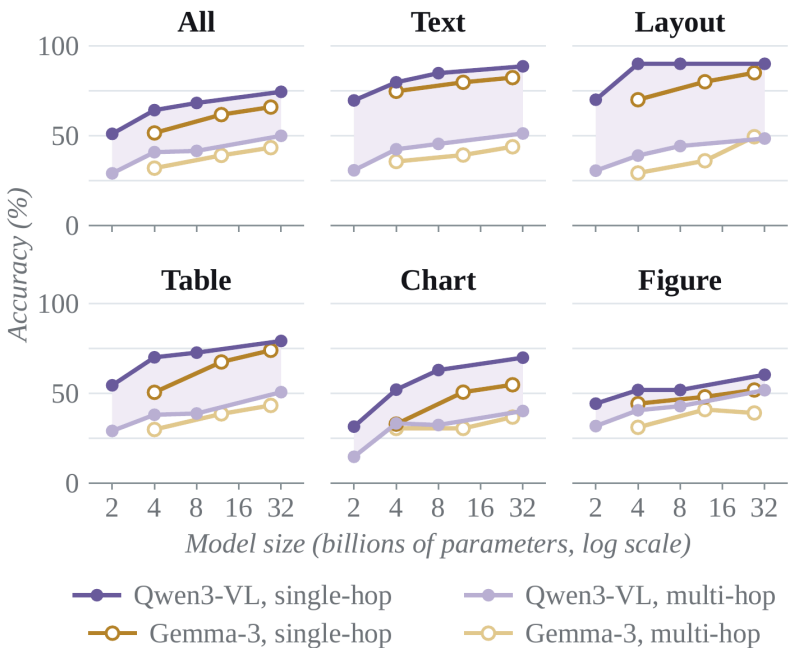
| Evidence coverage                                                                                                                                                                                                                                                                                                                                                                  | author list on pages 13, 14 | Distractor exposure                                                                                                                                                                                                                                                                                                                                                                                                                                | 10 gold pages of a 20-page deck |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <p>Q How many chapter authors of this report come from Columbia University?</p> <p>Gold 3</p> <p>page 13 ✗ “two” (Horton, Kinney)</p> <p>page 14 ✗ “one” (Garcia-Pando)</p> <p>--- ▼ add the complementary page -----</p> <p>both pages ✓ “three” (Horton, Kinney, Garcia-Pando)</p> <p>Each page is relevant, yet neither suffices: the answer is a property of the page set.</p> |                             | <p>Q How many lateral bar graphs are present in the document?</p> <p>Gold 10</p> <p>+3 pages ✗ “there are 12 lateral bar graphs”</p> <p>+5 pages ✗ “there are 13 lateral bar graphs”</p> <p>--- ▼ remove competing pages, or read on -----</p> <p>gold pages ✓ “there are 10 lateral bar graphs”</p> <p>+7 to +18 ✓ “there are 10 lateral bar graphs”</p> <p>A few added pages inflate the count; the gold set, or more context, counts right.</p> |                                 |

|            |                                                                                                                                                                                                            |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DESIGN     | — Prioritise <b>collective evidence coverage</b> while controlling distractors: structured navigation, summary-based reranking, or iterative retrieval.                                                    |
| DEPLOYMENT | — ColQwen3 lifts recall at 5 from <b>58.1% to 81.0%</b> over BM25, but needs a GPU and 110 ms per query against 0.3 ms. Judge retrieval depth by <b>end-to-end answers and latency</b> , not recall alone. |

# Reasoning: Results

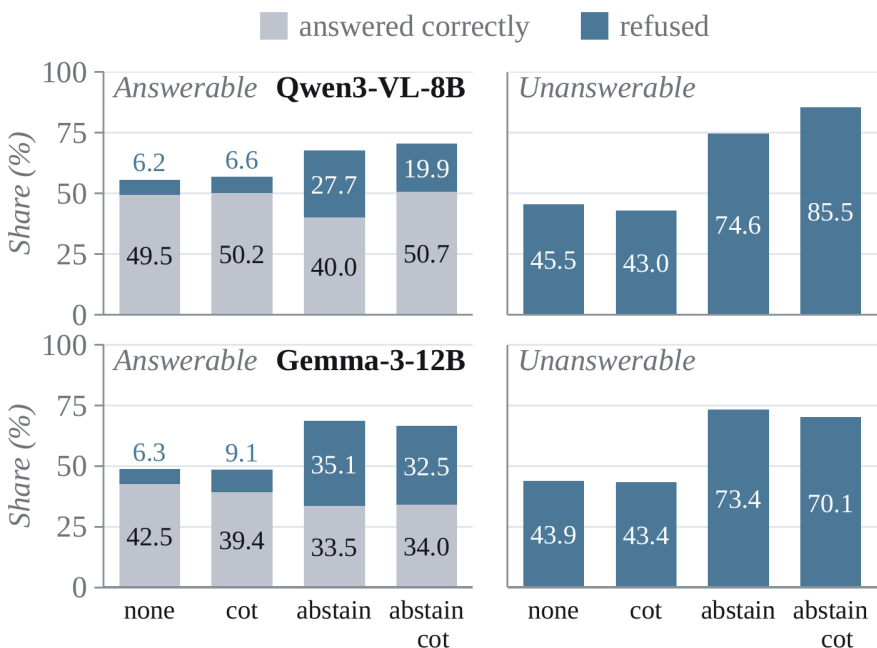
Is the evidence used correctly?

## Evidence integration



From 2B to 32B, multi-hop accuracy rises from **29.0%** to **49.9%**, but the gap to single-hop stays.

## Response calibration



An abstain instruction lifts correct refusal to 74.6%, but false refusal goes from **6.2%** to **27.7%**. Reasoning first recovers part of it.

# Reasoning: Design and Deployment

Is the evidence used correctly?

## Evidence integration

*a diagram, an ablation table*

**Q** If I drop the green module shown in Figure 6, what is the absolute percentage drop on 100-shot TACREV setting?

**Gold** 0.5%

**2B** ❌ “2.4%” (reads another table)

**4B** ❌ “-9.6%” (right table, wrong row)

**8B** ❌ “ $(75.9 - 74.9) / 75.9 \dots \approx 1.32\%$ ”

--- ▼ **a larger reasoner** -----

**32B** ✅ “ $75.9 - 75.4 = 0.5$ ”

*Each size breaks a different step of one chain: locate the table, read the cell, compute.*

## Response calibration

*two tables, two pages*

**Q** How much lower is the accuracy of GPT-4 on SituatedQA compared to MMLU Chemistry?

**Gold** 16.7%

**abstain** ❌ “Not answerable” (false refusal)

--- ▼ **reason before deciding** -----

**abstain\_cot** ✅ “ $79.9\% - 63.2\% = 16.7\%$ ”

**none** ✅ “... 16.7 percentage points lower”

**cot** ✅ “Step 1: ... Answer: 16.7%”

*The question is answerable; the abstain instruction turns multi-step difficulty into refusal.*

### DESIGN

- **Match capacity and inference strategy to the operation.** Separating answer construction from verification helps, at the cost of extra computation.

### DEPLOYMENT

- Quantisation keeps accuracy at a **smaller footprint but is not faster**: under a memory limit prefer a quantised larger model, under a latency limit fewer reasoning steps or a faster backbone.



# Conclusion

---

## SURVEY

### Organise

Multi-page document understanding is a problem of **evidence management**.

---

## EMPIRICAL STUDY

### Locate

Failures sit at **representation, selection, or reasoning**, and each needs a different fix.

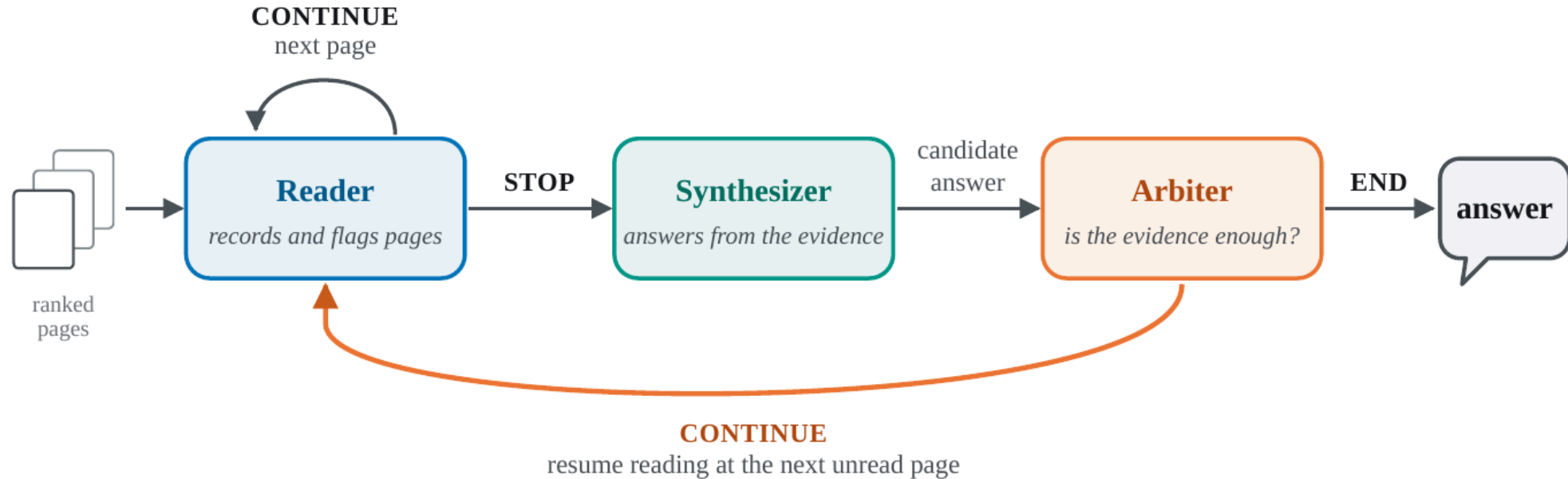
---

**Thank you** Questions?

# *Table of Contents*

- Introduction
- Survey of MP-VRDU
- Empirical Attribution Study
- Conclusion
- **Proof of Concept: A Simple Method**

# From Findings to a Simple Method



## REPRESENTATION

Every page is read as **text and image together**.

## SELECTION

**Read broadly**, then pass on compact records, not full pages.

## REASONING

A separate arbiter decides **whether the evidence is enough**.

# Simple Method Results

| Method                        | OCR | Trained | MMLongBench-Doc |             |             |                   | LongDocURL  |             |             |             |
|-------------------------------|-----|---------|-----------------|-------------|-------------|-------------------|-------------|-------------|-------------|-------------|
|                               |     |         | SIN             | MUL         | UNA         | ACC               | UND         | REA         | LOC         | ACC         |
| Gold pages + VLM              | ✓   | ×       | 57.7            | 37.4        | 52.5        | 49.5              | 69.7        | 63.6        | 59.5        | 65.7        |
| Whole document (via DocTrace) | ×   | ×       | 46.9            | 35.3        | 25.8        | 38.5              | —           | —           | —           | 45.1        |
| M3DocRAG (via SLEUTH)         | ×   | ×       | —               | —           | 48.6        | 45.9 <sup>a</sup> | 60.1        | 52.4        | 45.5        | 54.6        |
| DocAgent (via MARDoc)         | ✓   | ×       | —               | —           | —           | 46.6              | —           | —           | —           | —           |
| MDocAgent (via SLEUTH)        | ✓   | ×       | —               | —           | 49.7        | 47.8 <sup>a</sup> | 57.5        | 51.5        | 44.7        | 53.1        |
| MoLoRAG (via SLEUTH)          | ×   | ×       | —               | —           | 51.7        | 48.8 <sup>a</sup> | 63.4        | 52.7        | 49.3        | 57.6        |
| SLEUTH                        | ×   | ×       | —               | —           | 67.4        | 52.8 <sup>a</sup> | 65.7        | 53.0        | <b>53.6</b> | 60.0        |
| DocTrace                      | ✓   | ✓       | 53.2            | <b>41.1</b> | 70.5        | 52.9              | —           | —           | —           | 56.4        |
| <b>Ours (4-bit)</b>           | ✓   | ×       | <b>60.5</b>     | 39.2        | <b>76.9</b> | <b>56.5</b>       | <b>67.2</b> | <b>66.3</b> | 47.9        | <b>61.3</b> |

- **56.5%** on MMLongBench-Doc and **61.3%** on LongDocURL: highest among the Qwen3-VL-8B systems, training-free, at 4-bit.
- Multi-page questions still trail DocTrace, **39.2 against 41.1**.

# Inference-Time and Training Strategies

Three evidence-management decisions. Each can be **prompted around a frozen MLLM**, or **trained into it**.

## Retrieval and navigation

TRAINING-FREE

Similarity-based or relation-aware retrieval

TRAINED

Modality-alignment, cross-page-fusion, logical-relevance

## Reasoning strategies

TRAINING-FREE

Chain-of-Thought, Self-Reflection, Self-Consistency

TRAINED

Trace-imitation, then reward-driven (GRPO)

## Agentic methods

TRAINING-FREE

ReAct and tool use, multi-agent orchestration

TRAINED

Trajectory-distillation, reinforcement-learned

# Retrieval and Navigation

Retrieval **narrows a long document to compact evidence units** (chunks, pages, or graph nodes) for downstream reasoning.

- **Similarity-based retrieval.** Scores each evidence unit independently against the query and returns the top-ranked subset. Sparse lexical signals such as BM25 are the baseline; dense retrieval uses vision-language late-interaction encoders such as ColPali and ColQwen, often with an LLM reranking stage after.
- **Relation-aware retrieval.** Retrieves along explicit document structure. Tree-based variants follow section hierarchies and outlines, so chunks keep their parent context; graph-based variants traverse typed edges between headings, entities, and cross-references.
- **The trade-off.** Independent scoring handles paraphrase and cross-modal queries, but cannot follow cross-page references or multi-hop dependencies. Relation-aware methods can, at the cost of parser dependence and traversal cost.

Evidence missed or poorly ranked at retrieval is **unrecoverable at generation**. What is retrieved still has to be reasoned over.

# Reasoning Strategies

Reasoning strategies act over a **single evidence buffer assembled before the reasoning call**: the full document representation, a retrieved subset, or evidence from an agentic step. None updates model parameters.

- **Sequential reasoning: Chain-of-Thought.** Extends a single reasoning thread. CoT binds evidence from different pages into one chain and decomposes multi-hop questions into single-hop steps.
- **Sequential reasoning: Self-Reflection.** The model evaluates its own draft answer and revises it across rounds, optionally against a verifier or critic.
- **Parallel exploration.** Several reasoning paths over the same buffer. Self-Consistency samples multiple chains and takes the majority answer; sampling-adjudication replaces the vote with an LLM judge. Cost scales with the number of samples. Tree-of-Thoughts has seen little uptake.

A chain can only reason over the evidence already assembled. It **cannot recover what retrieval missed**.

# Agentic Methods

Agentic methods put one or more **LLM controllers** in charge: either iterating evidence acquisition under inference-time control, or coordinating across specialised roles.

- **ReAct and tool-augmented reasoning.** A single controller interleaves thought, action, and observation. Actions are document-side tool calls (retrievers, parsers, OCR engines) or structured navigation over a pre-built document representation.
- **Multi-agent orchestration.** Modality-specialised decomposition gives text and visual evidence separate controllers plus a synthesis controller. Role-specialised decomposition splits planning, execution, verification, and synthesis.
- **What the adaptive trajectory buys.** Compute scales with question difficulty rather than document length, the single-step recall ceiling of retrieval-augmented pipelines is lifted, and the action sequence is an inspectable audit trail.

The cost is **controllability**: inference cost is hard to bound, and a controller can commit early to wrong evidence and continue down the wrong trajectory.



# Training Strategies

Most systems adopt a backbone **already pretrained on single-page data**, so fine-tuning carries the weight. Training internalises the inference-time evidence-management decisions into model parameters.

## Trained retrieval

**Modality-alignment** aligns visual page embeddings with OCR-derived text. **Cross-page-fusion** adds neighbouring-page context to each page representation. **Logical-relevance** ranks pages by their position in a cross-page reasoning path.

RETRIEVER

## Trained reasoning

**Trace-imitation** supervises the backbone on demonstration chains in a fixed schema. **Reward-driven** training then refines it with GRPO, rewarding answer accuracy, evidence-page recall, and format compliance.

REASONER

## Trained agentic methods

**Trajectory-distillation** clones a stronger teacher's think-action-observation turns through SFT. **Reinforcement-learned** variants optimise the policy directly against trajectory-level rewards.

CONTROLLER

Two techniques recur across all three: **parameter-efficient tuning** (LoRA, adapters) and **multi-stage curricula** that go from single-page to document-scale supervision. All of it depends on the available datasets.

# Datasets

Datasets serve three roles, and the multi-page setting makes all three harder: **per-page annotation does not scale** to documents of tens or hundreds of pages, and long PDFs are harder to source and license than single pages.

## Pretraining

The exception rather than the norm. Confined to a few page-to-document and MLLM-centric systems that do continued pretraining on an existing backbone, because multi-page corpora are still small.

RARE

## Fine-tuning

Has converged on LVLM-synthesised supervision over scraped PDFs, with reinforcement learning increasingly layered on top to shape evidence-grounding and trajectory behaviour.

LVLM-SYNTHESISED

## Benchmarking

Three tiers: single-page extractive baselines, mid-length multi-page suites, and long-document multimodal benchmarks that need cross-page integration. The same small pool of PDFs gets reused.

CONTAMINATION RISK

**Answer-only accuracy masks retrieval failure** and rewards correct answers reached through incorrect reasoning. Newer benchmarks score retrieval and page-localisation as well, but it is not yet standard.

# Datasets and Evaluation

---

## PRIMARY

### MMLongBench-Doc

|            |             |            |              |
|------------|-------------|------------|--------------|
| <b>135</b> | <b>48.3</b> | <b>847</b> | <b>244</b>   |
| documents  | mean pages  | answerable | unanswerable |

- Annotated with **evidence pages, evidence sources and answerability**, which the experiments need.
- 367 of the answerable questions need two or more pages.

---

## COMPLEMENTARY

### LongDocURL

|            |             |              |
|------------|-------------|--------------|
| <b>396</b> | <b>85.6</b> | <b>2,325</b> |
| documents  | mean pages  | questions    |

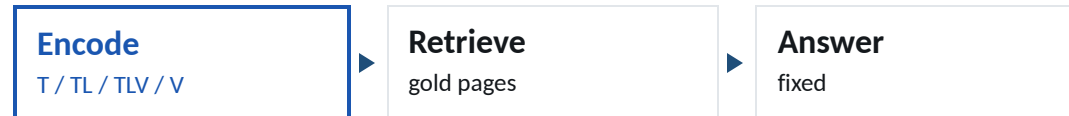
- **Born-digital only**, so answers are often in the text layer, even for figure questions.
- Used for the modality-ceiling result only: its evidence pages are not a minimal set.

---

**Judge:** Gemini-2.5-Flash at temperature zero returns correct, incorrect or abstained. Agreement is **96.0%** with GPT-4o and **98.3%** with a human annotator.

# Modality Ceiling

| Evidence               | Abstention  |             |            |            | Accuracy |             |             |      |
|------------------------|-------------|-------------|------------|------------|----------|-------------|-------------|------|
|                        | T           | TL          | TLV        | V          | T        | TL          | TLV         | V    |
| <b>MMLongBench-Doc</b> |             |             |            |            |          |             |             |      |
| Text                   | 23.0        | 11.5        | 7.3        | 8.0        | 36.9     | 44.4        | <b>51.0</b> | 42.3 |
| Layout                 | 20.3        | 10.2        | 0.8        | 4.2        | 26.3     | 33.9        | <b>49.2</b> | 43.2 |
| Table                  | 15.9        | 7.2         | 6.7        | <b>9.1</b> | 35.6     | <b>44.7</b> | <b>44.7</b> | 34.1 |
| Chart                  | 32.6        | <b>28.1</b> | 7.3        | 6.2        | 16.3     | 16.9        | <b>42.7</b> | 40.4 |
| Figure                 | <b>36.6</b> | 24.8        | <b>7.6</b> | 7.9        | 14.5     | 22.1        | <b>42.4</b> | 39.0 |
| <i>All</i>             | 26.3        | 17.2        | 6.2        | 7.1        | 29.0     | 35.7        | <b>49.5</b> | 42.6 |
| <b>LongDocURL</b>      |             |             |            |            |          |             |             |      |
| Text                   | 4.6         | 3.5         | <b>1.5</b> | <b>1.6</b> | 71.8     | 75.6        | <b>78.2</b> | 70.0 |
| Layout                 | 2.1         | 1.6         | 0.7        | 0.9        | 45.0     | 49.2        | <b>51.3</b> | 49.3 |
| Table                  | 5.0         | 3.2         | 1.1        | 0.8        | 57.7     | <b>65.6</b> | 63.6        | 52.4 |
| Figure                 | <b>6.7</b>  | <b>8.6</b>  | <b>1.5</b> | 1.3        | 59.9     | 62.8        | <b>69.9</b> | 61.9 |
| <i>All</i>             | 4.7         | 4.3         | 1.3        | 1.3        | 60.3     | 64.5        | <b>67.3</b> | 59.4 |



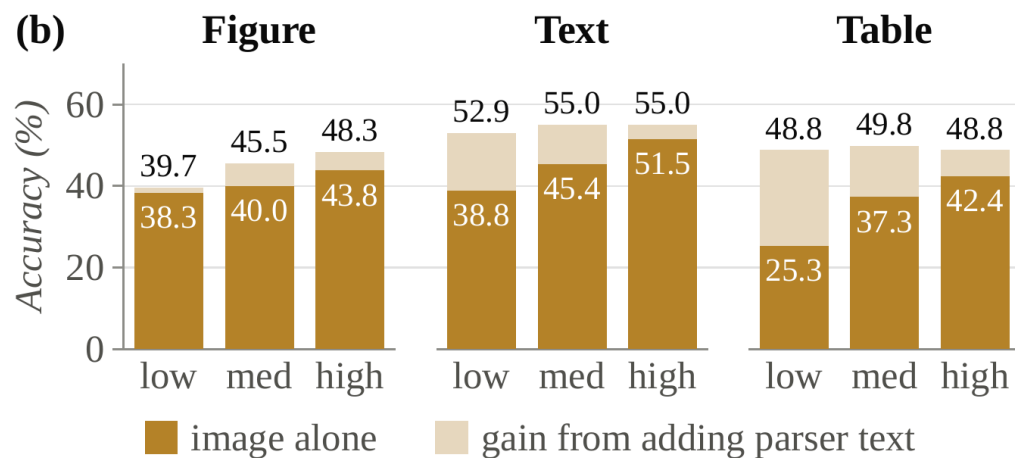
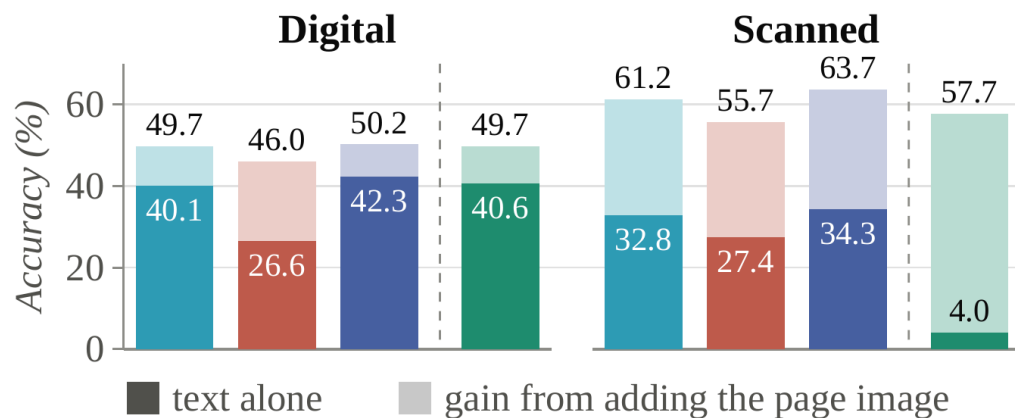
## REPRESENTATION

### Evidence accessibility depends on modality

- On MMLongBench-Doc, text-only input refuses **chart and figure** questions most. Page images bring refusal **under 10%**.
- Images alone stay below text plus images: 42.6% against 49.5%.
- On born-digital LongDocURL, adding vision to parser text gives only **2.8 points** overall, 7.1 on figure questions.

# Conversion Fidelity

(a) PaddleOCR MinerU Unlimited-OCR PyMuPDF



Encode  
parser, resolution

Retrieve  
gold pages

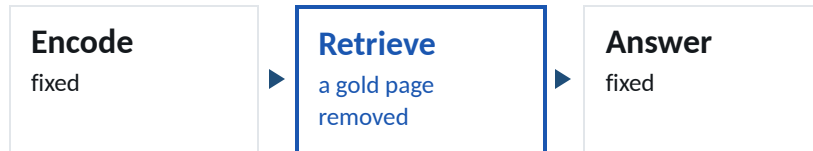
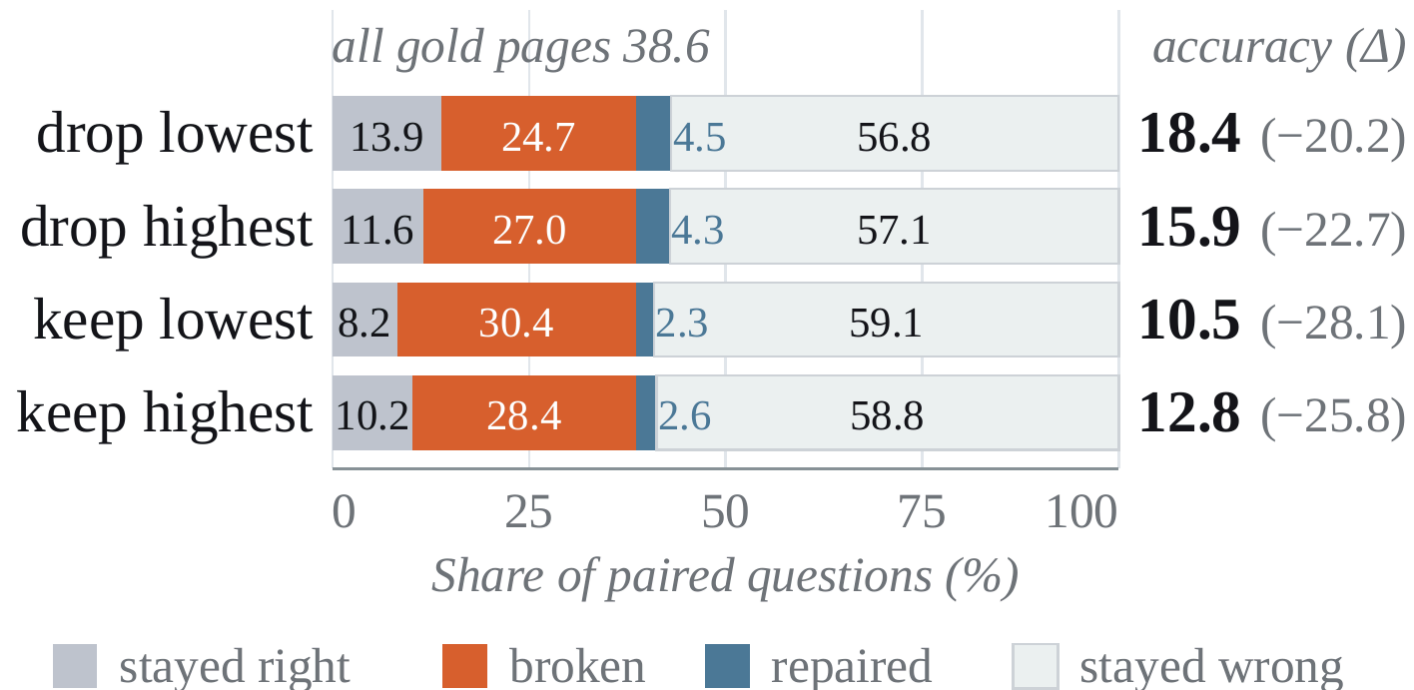
Answer  
fixed

REPRESENTATION

## Complementary modalities reduce conversion sensitivity

- Adding page images cuts the spread across parsers from **15.7 to 4.2 points** on digital documents.
- Higher resolution helps images alone far more than text plus images.
- Scanned documents stay sensitive to the parser, and figure questions to resolution.

# Evidence Coverage

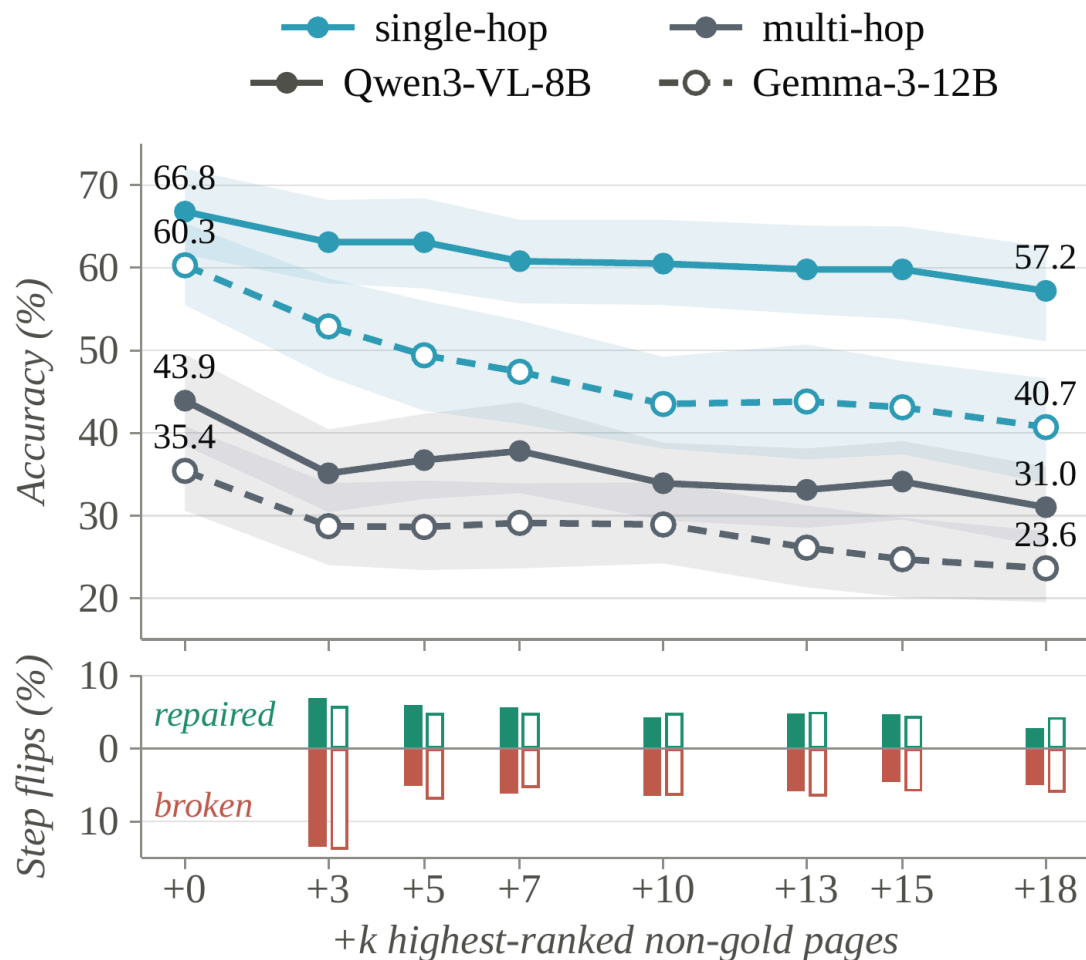


## SELECTION

### Omission causes large accuracy losses

- Removing **one gold page** takes accuracy from **38.6% to 18.4%** (lowest-ranked) or 15.9% (highest-ranked).
- Correct answers break far more often than wrong ones are repaired.
- Retrieval rank does not predict which page is safe to lose.

# Distractor Exposure



Encode  
fixed

Retrieve  
gold + k other pages

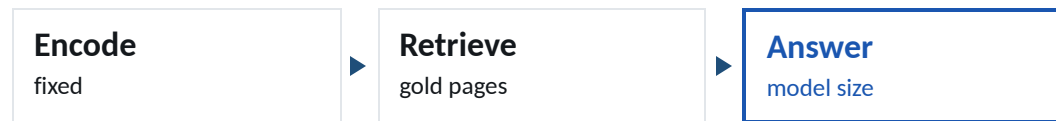
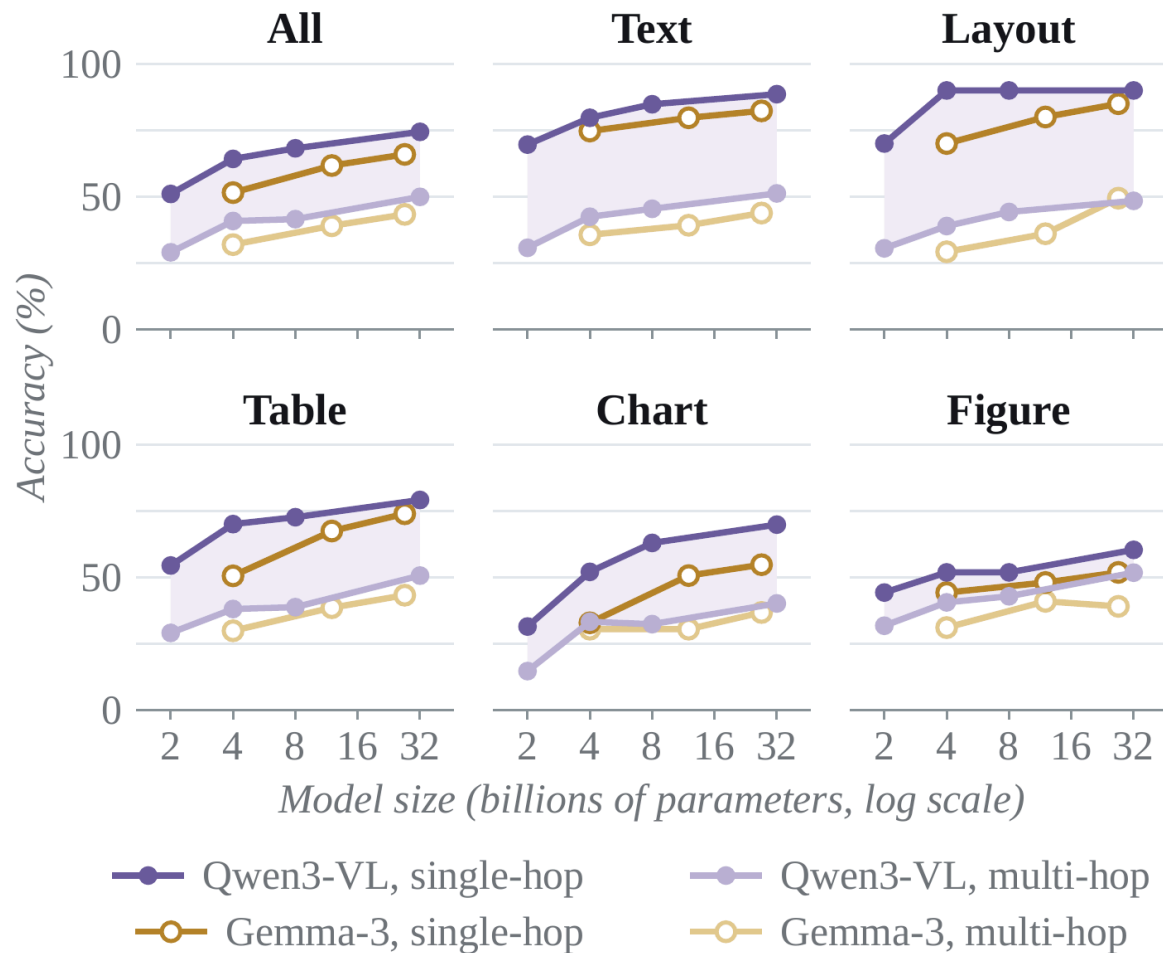
Answer  
fixed

## SELECTION

### Multi-hop accuracy drops sharply with added context

- Qwen3-VL-8B: multi-hop falls from **43.9% to 35.1%** after three extra pages. Single-hop declines gradually, 66.8% to 57.2%.
- Gemma-3-12B shows the same trend: multi-hop 35.4% to 23.6%, single-hop 60.3% to 40.7%.
- Every gold page is still there. Extra context both breaks and repairs answers.

# Evidence Integration



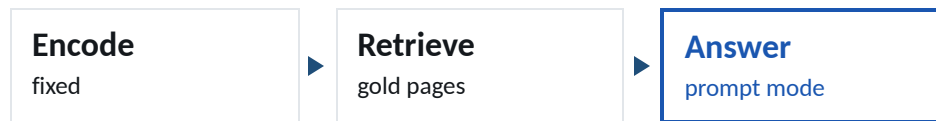
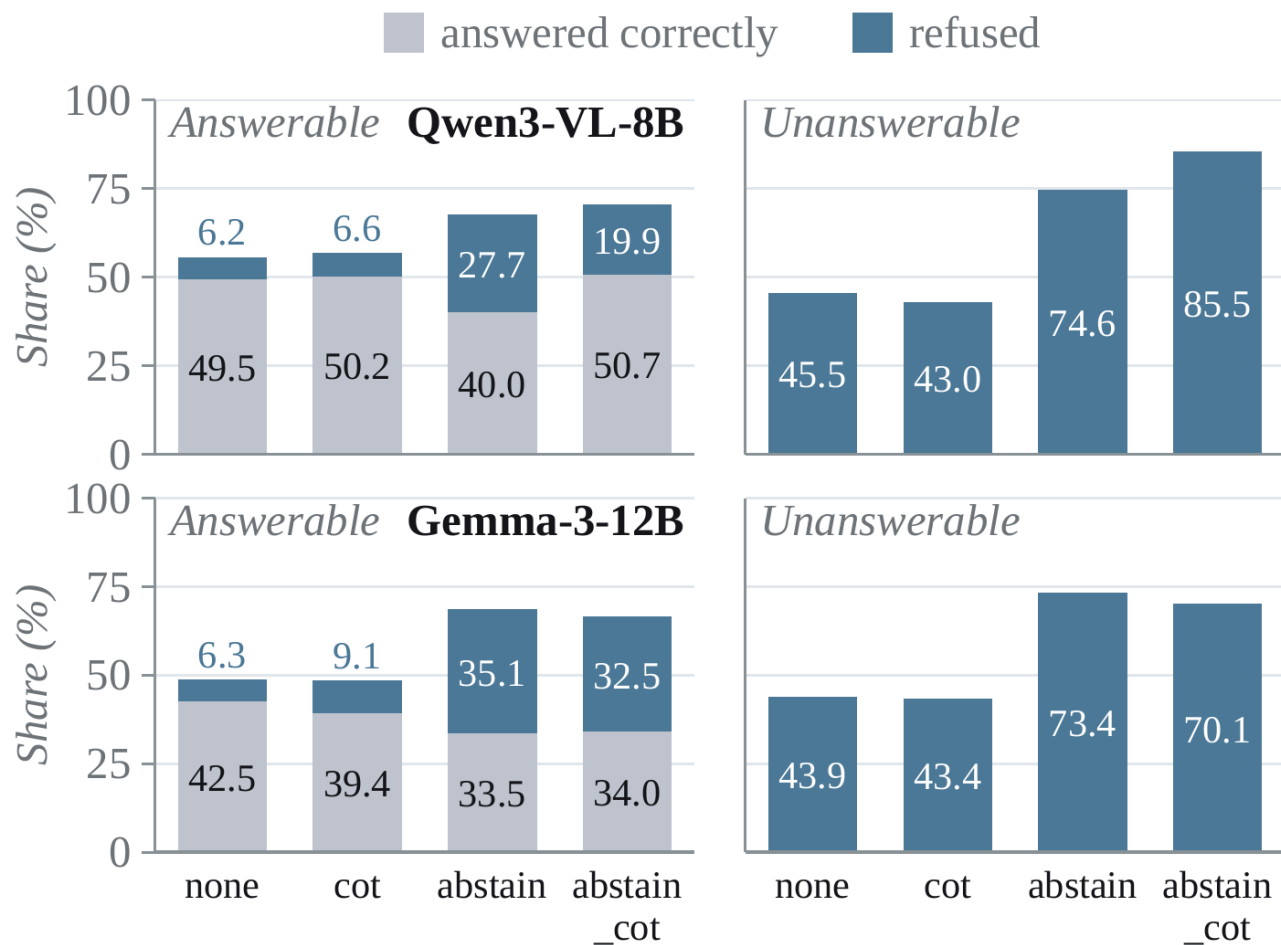
## REASONING

### Scaling helps, but the multi-hop gap persists

- Qwen3-VL from 2B to 32B: multi-hop rises from **29.0% to 49.9%**, single-hop from 51.0% to 74.4%.
- The gap between them stays roughly the same.
- Gemma-3 shows the same separation.



# Response Calibration



## REASONING

### Reasoning before abstaining limits false refusal

- Qwen3-VL-8B, abstain: unanswerable refusal **45.5% to 74.6%**, but false refusal **6.2% to 27.7%** and accuracy 49.5% to 40.0%.
- Adding CoT: false refusal 19.9%, accuracy 50.7%, unanswerable refusal 85.5%.
- Gemma-3-12B gains much less from CoT: false refusal 35.1% to 32.5%.

# Deployment: Accuracy and Cost

| Representation      | Acc. | Text | Vis. | Tokens   | Prefill   | Decode   |
|---------------------|------|------|------|----------|-----------|----------|
| T                   | 29.0 | 499  | 0    | 499      | 1.4       | 5.7      |
| TL                  | 35.7 | 874  | 0    | 874      | 1.8       | 6.6      |
| TV                  | 50.1 | 500  | 459  | 959      | –         | –        |
| TLV                 | 49.5 | 874  | 459  | 1333     | 26.3      | 6.7      |
| V                   | 42.6 | 20   | 459  | 479      | 29.0      | 7.1      |
| Retriever           | R@1  | R@5  | R@10 | Idx. (s) | Qry. (ms) | GB       |
| BM25                | 32.7 | 58.1 | 68.1 | 0.02     | 0.3       | CPU      |
| BGE-M3              | 32.8 | 61.3 | 72.4 | 0.01     | 47        | 1.2      |
| Qwen3-Emb.          | 33.5 | 63.9 | 74.8 | 0.19     | 89        | 13.1     |
| ColModernVBERT      | 47.1 | 74.0 | 82.7 | 0.91     | 41        | 5.1      |
| ColQwen2.5          | 49.2 | 77.2 | 84.8 | 1.50     | 293       | 8.6      |
| ColQwen3            | 53.7 | 81.0 | 87.9 | 0.57     | 110       | 9.4      |
| Reasoner            | T    | TL   | TLV  | V        | GB        | Time (s) |
| Qwen3-VL-8B, bf16   | 31.9 | 38.4 | 52.4 | 45.5     | 17.5      | 30.7     |
| 8-bit               | 32.5 | 37.9 | 53.6 | 45.7     | 10.3~     | 43.5     |
| 4-bit               | 31.8 | 39.4 | 51.8 | 45.5     | 6.8~      | 35.4     |
| Qwen3-VL-32B, 4-bit | 37.0 | 42.1 | 59.1 | 52.3     | 20.0~     | 47.5     |
| InternVL3-8B        | 19.3 | 25.3 | 32.6 | 24.1     | 15.9      | 5.3      |
| MiniCPM-V-4.5       | 30.7 | 38.1 | 60.1 | 53.3     | 17.4      | 47.2     |
| GLM-4.6V-Flash      | 25.9 | 32.9 | 42.7 | 30.9     | 20.6      | 19.8     |
| Llama-3.2-11B-V     | 26.5 | 34.3 | 33.3 | 25.3     | 21.3      | 10.4     |
| Gemma-3-12B         | 30.5 | 36.9 | 48.1 | 41.6     | 24.4      | 8.0      |

## PRACTICAL DISCUSSION

### Every choice has a cost side

- **Representation:** embedded text plus images matches parser text plus images (50.1% against 49.5%) with fewer tokens. Images cost about 26 s of prefill against 1.8 s.
- **Retriever:** ColQwen3 reaches 81.0% recall at 5 against 58.1% for BM25, but needs a GPU and 110 ms per query against 0.3 ms.
- **Reasoner:** 4-bit keeps accuracy at a smaller footprint (51.8% in about 6.8 GB) but is not faster.

# From Findings to Design

The survey and the experiments both point at design principles. I put them together in a **deliberately simple system**, to see whether they hold up at the level of a whole pipeline rather than one controlled condition.

## REPRESENTATION

### Keep text and images together

- ▶ Every page read at the full TLV representation, with a strong parser and high-resolution rendering, since the two channels cover for each other's failures.

## SELECTION

### Gather broadly, present selectively

- ▶ Missing evidence is unrecoverable and moderate distraction is not, so read widely in retrieval-ranked order, but carry most of it forward as compact records rather than as full pages.

## REASONING

### Check whether the answer is supported

- ▶ Because the reasoner cannot judge sufficiency on its own, a separate step decides whether to accept the answer or keep reading.

No planner, no tool set, no query reformulation. **The point is to validate the principles, not to introduce a complex architecture.**

# Simple Method Cost

|                      | Turns | Tokens     |              |        | Latency (s) |        | Peak<br>VRAM (GiB) |
|----------------------|-------|------------|--------------|--------|-------------|--------|--------------------|
|                      |       | Input text | Input visual | Output | Prefill     | Decode |                    |
| reader               | 11.5  | 5,030      | 816          | 697    | 1.8         | 46.8   | 7.2                |
| synthesizer          | 1.2   | 9,307      | 2,018        | 463    | 3.8         | 34.7   | 8.1                |
| arbiter              | 1.2   | 6,806      | 1,734        | 464    | 2.9         | 33.3   | 7.9                |
| Per question, mean   | 13.8  | 76,669     | 13,767       | 9,092  | 28.5        | 617.5  | 8.2                |
| Per question, median | 9.0   | 43,113     | 9,830        | 6,078  | 15.8        | 385.0  | 8.0                |

- **13.8 turns** per question on average, 11.5 of them the reader.
- The cost is time, not memory: **10.8 minutes** mean, 6.7 median, with peak memory at 8.2 GiB.